NIKHIL GOUD PALAMAKULA

Full-Stack Developer | React · TypeScript · Node.js · NestJS · MongoDB

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PROFESSIONAL SUMMARY

Full-Stack Developer with 5+ years of engineering experience, including 3+ years building production web applications on React, TypeScript, Node.js, NestJS and MongoDB. Experienced across the full software development lifecycle — requirements, database schema design, REST API development, frontend engineering, performance optimisation and deployment support. Shipped multi-tenant SaaS products, enterprise ERP systems, real-time dashboards and role-based B2B applications from the ground up. Served as technical lead on an enterprise ERP, owning data architecture, API design and coding standards across the codebase. Comfortable owning features end to end and working directly with stakeholders to turn business requirements into shipped software.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, SQL, HTML5, CSS3, SCSS

Frontend: React 19, Redux, React Hooks, Material UI (MUI), Ant Design, Vite, Highcharts, Responsive Web Design, REST API integration

Backend: Node.js, NestJS, Express.js, REST API design, Microservice-style modular architecture, JWT authentication, Role-Based Access Control (RBAC), WebSockets (Socket.IO), Cron jobs, Server-side PDF generation (EJS)

Databases: MongoDB, Mongoose, SQL, Redis, Schema design, Data modelling, Indexing, Query optimisation, Aggregation pipelines

Cloud & DevOps: Docker, Docker Compose, AWS, Git, GitLab, CI/CD pipelines, Nginx

Practices: Agile delivery, Code review, Coding standards, Multi-tenant architecture, Performance optimisation, Data architecture, Technical leadership

Additional (Industrial Domain): PLC programming (Mitsubishi, Allen-Bradley), SCADA, HMI, 21 CFR Part 11 compliance, Audit trails, IoT telemetry

KEY ACHIEVEMENTS

- **Co-led delivery** of an enterprise ERP as one of two technical leads in a 6-person team; owned data architecture, schema design and coding conventions across a ~89-module backend serving ~200 daily users.
- **Sole full-stack developer** on a B2B analytics platform — 45+ REST API modules and a 40+ module React frontend serving four distinct user roles.
- **Reduced recurring data-fetching cost by ~$1,100** by eliminating redundant data fetches and caching results so a single fetch serves multiple downstream calculations.
- **Engineered real-time features** with WebSockets (Socket.IO), streaming authenticated live updates to dashboards with per-tenant connection isolation.
- **Hardened application security** with JWT authentication, refresh tokens, role-based access control and scoped service tokens across multi-tenant products.
- **Improved API response times** by ~50% (approx. 3s to 1.5s) through database indexing, query tuning and pagination on high-volume analytics endpoints.

PROFESSIONAL EXPERIENCE

Full-Stack Developer | PatternLab.ai | May 2024 – Present

Building multi-tenant B2B SaaS platforms for manufacturing, energy and export businesses using React, TypeScript, NestJS and MongoDB.

- Co-led SeaTrace, an enterprise processing and export ERP, as one of two technical leads in a 6-person engineering team — owning schema design per module, data storage and fetch patterns, and codebase conventions; the platform now supports ~200 daily users.
- Architected and delivered the exports and finance domain end to end: export masters and transactions with server-side PDF document generation, a strategy-pattern payments module, and raw-material payments-outstanding tracking.
- Developed features across processing, quality-control and reporting modules in a NestJS + MongoDB backend (~89 REST modules) and a React 19 + TypeScript + Vite + MUI frontend.
- Redesigned downtime analytics for a multi-tenant platform serving 6 client organisations, making idle machine segments count toward stoppage aggregation and converting unexplained gaps into attributed, reportable data.
- Engineered real-time state visibility with an authenticated Socket.IO gateway, JWT-over-cookie handshake and a per-tenant connection registry, pushing live updates to operational and management dashboards.

- Contributed to the Python analytics engine for data processing and KPI calculation; co-designed the MongoDB data model and cut recurring data-fetching cost by ~$1,100 through caching and fetch consolidation.
- Scaled a strategy-pattern reporting engine to 15 report types, generating server-side PDFs via EJS and Highcharts Export Server on cron schedules.
- Owned an Energy Management platform as sole full-stack developer: MongoDB schema design, 45+ REST API modules and a 40+ module React frontend serving four user roles.
- Secured platform access with JWT authentication, refresh tokens, bcrypt password hashing, automatic session expiry and role-based access control.
- Cut API response times by ~50% (approx. 3s to 1.5s) through database indexing, query tuning and pagination; containerised a three-service stack with Docker Compose for on-premise deployment.

Full-Stack Developer | Matrical Technologies | Jun 2023 – May 2024

- Developed a Hospital Management System spanning the end-to-end patient journey from walk-in registration to discharge — 4 core modules covering patient records, billing, pharmacy dispensing and inventory — using React, Node.js/Express and MongoDB.
- Coordinated directly with clients to gather and clarify requirements, translating business needs into technical specifications and delivering features against them.
- Demonstrated the platform to existing and prospective clients, gathering feedback for product improvements and supporting pre-sales pitches that brought in new business.

SCADA Engineer | SR Automation | Jan 2022 – Jun 2023

- Designed SCADA interfaces and worked with SQL for industrial automation systems across pharmaceutical manufacturing plants.
- Liaised directly with plant and client engineering teams to capture requirements for new machine functionality and process changes, then specified, built and commissioned the automation on site.
- Executed automation and 21 CFR Part 11 compliance projects across 4 pharmaceutical companies — Hetero, MSN, Aurobindo and Dr. Reddy's — commissioning Beijer X2 Pro HMI systems in place of legacy Mitsubishi and Allen-Bradley HMIs and PLCs, often running several plant units in parallel.
- Introduced user management, audit trails and electronic reporting to satisfy regulatory requirements; reworked machine process-flow logic per plant while preserving core process structure.
- Engineered injectables mixing-vessel automation from scratch at Eugia Unit 3 — Mitsubishi PLC logic, Beijer X2 Pro HMI, production and batch reporting, and audit trails.

Automation & Marketing Engineer | SR Automation | Nov 2020 – Dec 2021

- Programmed and commissioned PLC systems (Mitsubishi, Allen-Bradley) for manufacturing clients.
- Owned techno-commercial responsibilities alongside engineering delivery — client negotiations, new order generation and quotation-to-closing on projects including Mahindra & Mahindra.

SELECTED PROJECTS

SeaTrace — Enterprise ERP Platform | React, TypeScript, NestJS, MongoDB | app.seatrace.ai

- End-to-end ERP supporting ~200 daily users across procurement, operations, quality control, exports and payment settlement, with full audit traceability, configurable forms, role-based multi-site access, a 12-module material-balance system, 13 quality-inspection forms and 30+ analytical reports.

Factory Intelligence — Multi-Tenant SaaS Analytics Platform | React, TypeScript, NestJS, MongoDB, Python | app.patternlab.ai

- Multi-tenant SaaS platform live across 6 client organisations, combining device telemetry with user-entered data to compute operational KPIs, with real-time WebSocket dashboards, a 15-type reporting engine, feature flags, RBAC and an AI analytics copilot backed by a Python agent.

EMS — Energy Management Platform | React, Node.js, MongoDB, Python, Docker

- Monitoring and alerting platform with live consumption tracking, configurable threshold alerts, cost analytics, recurring email reporting and exportable CSV/PDF reports across four user roles.

EDUCATION

Master of Business Administration (MBA) | Jawaharlal Nehru Technological University (JNTUH), Hyderabad | 2025 – Present (Pursuing)

B.Tech, Electronics & Instrumentation Engineering | Jawaharlal Nehru Technological University (JNTUH), Hyderabad | 2016 – 2020